

WENTWORTH LIU

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Data Scientist · Predictive Modelling · User Segmentation · Decision Science

SUMMARY

Data Scientist with 2+ years of experience applying Python, SQL, statistical modelling, and machine learning to product and operational decisions. Built explainable risk-ranking, quasi-causal analysis, NLP classification, and reusable feature workflows for million-scale EV fleets at Li Auto; previously delivered multi-source measurement, user-segmentation, and content-prioritisation analytics at ByteDance. Strong at translating model outputs into actionable prioritisation and decision support across complex data environments.

EXPERIENCE

Li Auto Beijing, China
Data Scientist, Battery Intelligence Department Jul 2024 – Present

- **Predictive Risk Ranking & Decision Support.** Recast thermal-runaway detection under extreme class imbalance as an explainable ranking task; built a cross-platform ML pipeline generating vehicle-level risk scores and feature-level drivers across multiple battery platforms, achieving 93% Recall@2% on retrospectively validated high-risk cases while limiting review to the highest-risk 2% of vehicles.
- **Causal Analytics & Prioritisation.** Used quasi-causal analysis on million-scale fleet telemetry to identify high-risk SOH-degradation cohorts and actionable drivers; productised intervention analysis across multiple vehicle models, supporting targeted strategies projected to slow degradation and reduce warranty exposure by tens of millions of RMB for a single model.
- **Reusable Feature Workflows.** Built a configurable feature system for multi-dimensional telemetry, using custom event definitions, multi-operator aggregations, and cross-platform normalisation across multiple battery platforms; deployed reusable workflows to an internal ML platform and improved feature-iteration efficiency by ~85%.
- **NLP Classification & Review Optimisation.** Built a hybrid rule-based and statistical classifier for battery-quality work orders across 30+ fault categories, combining text matching, TF-IDF vectors, and Random Forest; achieved Weighted F1 > 0.90 and used confidence-based routing to focus review effort on ambiguous cases.

ByteDance Beijing, China
Data Analytics Intern, Douyin Market Strategy May – Sep 2023

- **Multi-source Measurement & Monitoring.** Integrated consumption, dissemination, and search signals across multiple data platforms into an IP popularity measurement system; added automated pipeline monitoring to address upstream data instability and support recurring content-investment and promotional decisions.
- **User Segmentation & Content Prioritisation.** Redesigned performance metrics and built a multi-dimensional content taxonomy for Qishui Music; diagnosed new-user conversion decline through funnel segmentation and KOL audience-saturation modelling, informing content selection and marketing-resource allocation.

Dongwu Securities China
Equity Research Intern, Mechanical Sector Dec 2021 – Mar 2022

- **Quantitative Financial Analysis.** Combined company fundamentals, competitive analysis, and market dynamics with stock-price modelling to deliver short-, medium-, and long-term investment recommendations for Harmonic Drive Systems.

EDUCATION

Nanjing University Sep 2021 – Jun 2024
MSc Computational Materials Science

- Developed Skeaf, a 3D Fermi-surface visualisation tool combining HDBSCAN clustering with quantum-oscillation data; first author, Physical Review B (2024); co-author, Physical Review B (2024) and Physical Review Materials (2023); Patent: CN115795266A.

Nanjing University Sep 2017 – Jul 2021
BSc Materials Science & Engineering (Outstanding Graduate) · GPA: 4.6/5.0 · Top 1% of cohort

- National Scholarship (top 0.15%) · MCM/ICM 2020 Honorable Mention (Team Captain) · Sakura Science Exchange Programme, Tohoku University, Japan (5 selected university-wide).

SKILLS

Programming	Python · SQL/HiveSQL · C++ · MATLAB
ML & Statistics	Machine Learning · Statistical Modelling · Predictive Modelling · Causal Inference · Model Evaluation · Feature Engineering · NLP · Time Series · LightGBM · scikit-learn
Data & Product	User Segmentation · Risk Ranking · Multi-source Data Integration · Monitoring · Data Productisation · Decision Support · Visualisation · Linux · Hadoop
Domain & Lang.	Product Analytics · EV Analytics · Financial Research · Scientific Computing · English (TOEFL 104) · Mandarin/Cantonese (Native) · Japanese (JLPT N1)